

Session 1 Problems

A. Absolute Subtraction

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Omar has a problem calculating the absolute subtraction of two numbers, since he can't recognize the bigger number. Your objective is to write a program which takes the value of two numbers and outputs the absolute difference between the two numbers.

Input

The input consists of two integers X and Y ($0 \leq |x|, |y| < 10^3$).

Output

Print one line containing the absolute (non-negative) value of the difference between the two numbers.

Sample test(s)

input
5 7
output
2

input
7 5
output
2

B. Integers and Decimals

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

Mohammad is a student in third grade, first lesson in math is an introduction to the integers and decimal numbers. His teacher gave him homework, it's about the difference between integer and decimal number. So, Mohammad ask you to help him with his homework.

Input

Given a number N where $(0 < N \leq 10^{12})$

Output

Print the word "Integer" if N is integer , and "Decimal" if N is decimal number.

Sample test(s)

input
10
output
Integer
input
9.8
output
Decimal
input
1000000.88
output
Decimal

C. Climbing Stairs

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

Sarah likes to climb up a stair of N steps. She can climb 1 or 2 steps at each move. Now,She wants to know What is the minimal number of moves making her climb to the top of the stairs .

Input

Given N where $(1 \leq N \leq 1000)$

Output

Print the answer in one line.

Sample test(s)

input
10
output
5
input
135
output
68
input
1
output
1

D. Leap of Faith

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

You may have played "Assassin's Creed" game before, but did you try to perform it yourself?! Excited?!

Before doing it you must realize the risks of the jump to decide performing it or not.

You will be given the height of the building in meters. You have to calculate your velocity at the moment you will hit the floor (m/s).

Input

You will be given an integer H ($1 \leq H \leq 1000$) the height of the building in meters.

Output

Print the impact velocity. The output should be rounded to three decimal points.

Sample test(s)

input
15
output
17.321

input
18
output
18.974

input
720
output
120.000

Note

- Assume that the acceleration of the gravity = 10 m/s^2 .

- You are initially standing on the top of the building.

- $(Velocity)^2 = (initialVelocity)^2 + 2 * G * H$ (m/s).

E. Esto3'omaya

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Esto3'omaya is an old Egyptian game consisting of 4 players where one of them counts from 1 to 10 then searches for the other 3 players and the last one of those 3 players he finds wins the game.

Given the time at which each player appears, determine who the winner is.

In this problem you will be given 3 integers each one represents the time (in minutes) that the player appears in, find the winner of the game.

Input

The input consists of 3 integers A, B, C. ($0 < A, B, C < 100$) and ($A \neq B \neq C$).

Output

Output on a line that consists of the number of the winner.

Sample test(s)

input
3 2 1
output
1

input
10 15 20
output
3

F. Geometry is fun!

time limit per test: 2.5 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

Ameen is a skilled problem solver who loves geometry. He has passed by this difficult problem that he needs your help with.

Ameen has a right triangle JKL that is right at angle K . A line has been drawn from angle K and was perpendicular to the side JL and meets JL at point M . This line represents the altitude of the triangle. The triangle looks like the following figure:[figure here](#)

Ameen has the length of lines JK and JM . He wants to find the length of ML . Help Ameen solve this problem.

Input

The first line of the input will be a single integer T , the number of test cases. Then T lines follow.

Each line contains 2 integers X and Y ($1 \leq X \leq Y \leq 10^4$) that represent the length of the lines JK and JM , respectively.

Output

The output should contain T lines, one for each test case.

For each test case, print a single real number that represents the length of ML . The answer of each test case is considered correct if the absolute error does not exceed 10^{-6} .

Sample test(s)

input
2 8 4 9 6
output
12 7.5

G. Recognize The Shape

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Ahmed has a quadrilateral with four right angles but he doesn't know if this shape is a square or a rectangle. He knows the area of the shape and he is sure that the side's lengths of this shape have integer numbers.

Please help Ahmed recognize the shape.

Input

The single line of the input contains integer A ($1 \leq A \leq 10^5$) which denotes the area of a shape.

Output

Print "Rectangle" if the shape is a rectangle, "Square" if the shape is a square, and if it can be both print "Unknown".

Sample test(s)

input
6
output
Rectangle

input
9
output
Unknown

Note

For the first sample the sides of this shape can be 1 X 6 or 2 X 3 and in both cases the shape is a rectangle.

For the second sample the sides of the shape can be 1 X 9 or 3 X 3 so it can be rectangle or square.

H. Mysterious Machine

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

"MUWAHAHAHAHAHAA, IT IS ALIVE!" Dr.Codenstein said in his lab. Dr.Codenstein has just invented a mysterious machine that no one knows what it does. Not even himself!

Trying to figure out what the machine does he gave it some numbers, "SUM...SUM..SUM..SUM" the machine said, then it gave back some other numbers, but the doctor can't figure out the relation between the input and output.

Can you find the relation and simulate the machine?

Input

Tests consist of a single integer N ($1 \leq N \leq 10^9$).

Output

Print the machine's response.

Sample test(s)

input
1
output
1

input
10
output
55

input
4
output
10

I. You Shall Not Pass

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

Ahmed is a first year student at faculty of computers and information . He has just finished his final exams , but he is worried and keeps estimating his grades.

Ahmed needs your help , he will give you his grades estimates and you should tell him he will pass or not.

Ahmed studies 6 subjects and to pass he has to get at least X marks in each subject and the total marks must be at least Y points.

Input

First line will consist of two integers X,Y ($1 \leq X, Y \leq 1000$). Second line will consist of six integers (≤ 1000) represent the marks Ahmed thinks he will get in each subject.

Output

Print one line with "Pass" or "Not Pass" (without the quotes).

Sample test(s)

input
5 50 10 10 10 10 10 10
output
Pass

input
5 100 10 10 10 10 10 10
output
Not Pass

J. Boring

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Have you ever heard about "Loops"?

"Loops" are a way that we use in programming to repeat some operations many times without writing the same code again and again.

To understand the concept of "Loops" you will have to solve this problem in an old boring way.

Input

Given 5 integers N ($1 \leq N \leq 10^9$).

Output

Print X Y - which X is the number of odd numbers, and Y is the number of even numbers.

Sample test(s)

input
1 2 3 4 5
output
3 2

input
10 100 1000 10000 100000
output
0 5